

Full symmetric monodromy throughout Gallagher’s weighted-lift family

An attributed Brink–Gallagher corollary

Alec Kriebel, with substantial AI assistance

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Status, priority, and review notice. Gallagher has priority for the weighted-lift maps and their generic degrees. Brink’s 2004 theorem supplies the abstract two-free-coefficient Galois calculation. A MathOverflow answer and linked Note 19 had already proved full symmetric monodromy for Gallagher’s canonical tower in generic degrees 3 through 13. The contribution recorded here is only the uniform all-degree, all-admissible-seed corollary obtained by combining Gallagher’s exact inverse equation with Brink’s theorem. A source audit found no earlier statement of that combination, but cannot guarantee worldwide priority. This work is not peer reviewed. Exact computations check encoded algebra; they are evidence against transcription errors, not peer review.

Abstract

Gallagher constructed dimension-three Keller counterexamples of every generic degree $d \geq 3$ by a weighted-lift procedure depending on an admissible polynomial seed. We observe that the inverse equation for every such seed is

$$\Phi(T) - PT + cQ = 0,$$

where Φ has degree d and P, Q are independent target parameters. Brink’s classical theorem on a polynomial with free linear and constant coefficients therefore applies verbatim. Exact rational recovery identifies the root field with the function-field extension of the Keller map and rules out extraneous denominator branches. Consequently every admissible weighted lift of generic degree d has geometric monodromy S_d . In particular every symmetric group S_d , $d \geq 3$, is realized as the geometric monodromy group of a Keller counterexample $\mathbb{A}_{\mathbb{C}}^3 \rightarrow \mathbb{A}_{\mathbb{C}}^3$.

1 Statement

For a dominant polynomial map $F : \mathbb{A}_{\mathbb{C}}^3 \rightarrow \mathbb{A}_{\mathbb{C}}^3$, write

$$K = \mathbb{C}(F_1, F_2, F_3), \quad L = \mathbb{C}(x, y, z).$$

Its generic degree is $[L : K]$, and its geometric monodromy group is the Galois group of the Galois closure of L/K , in its natural action on the generic fibre.

Theorem 1. *Let $p \in \mathbb{C}[w]$ be any admissible seed in Gallagher’s weighted-lift construction and suppose $\deg p = d - 1 \geq 2$. The associated Keller counterexample has generic degree d and geometric monodromy group S_d .*

Corollary 2. *For every $d \geq 3$, the symmetric group S_d occurs as the geometric monodromy group of a Keller counterexample $\mathbb{A}_{\mathbb{C}}^3 \rightarrow \mathbb{A}_{\mathbb{C}}^3$ of generic degree d .*

2 The weighted lift

Fix $b, c \in \mathbb{C}^\times$. An admissible seed is a polynomial p satisfying

$$p(0) = 0, \quad p(1) = -c, \quad \int_0^1 p(s) ds = 0, \quad \kappa := \frac{p'(1)}{c} \neq -2. \quad (1)$$

Define

$$q(0) = 0, \quad q'(w) = \frac{wp'(w)}{c}, \quad a = -\frac{1 + \kappa}{2 + \kappa}. \quad (2)$$

Put

$$v = xy, \quad t = x^2z, \quad u = 1 + v, \quad \gamma = 1 + av + bt, \quad w = u\gamma$$

and

$$\beta = c + \frac{p(w)}{\gamma}, \quad \alpha = u + \frac{q(w)}{\gamma^2}.$$

Gallagher's weighted lift is

$$F_p = (A, B, C) = \left(\frac{\alpha}{x^2}, \frac{\beta}{x}, x\gamma \right). \quad (3)$$

For completeness, the apparent divisions in (3) cancel. The condition $p(0) = 0$ makes $p(u\gamma)/\gamma$ polynomial in u, γ . Since $q(0) = q'(0) = 0$, the same holds for $q(u\gamma)/\gamma^2$. At $(v, t) = (0, 0)$ one has $u = \gamma = w = 1$. Integration by parts gives

$$q(w) = \frac{wp(w) - \Phi(w)}{c}, \quad \Phi(w) := \int_0^w p(s) ds, \quad (4)$$

so (1) gives $q(1) = -1$. Thus $\alpha(0, 0) = \beta(0, 0) = 0$, while

$$\frac{\partial \alpha}{\partial v}(0, 0) = 1 + \kappa(1 + a) + 2a = 0.$$

After substituting $v = xy$ and $t = x^2z$, β is divisible by x and α by x^2 .

We also recall the determinant calculation. Set

$$P = BC = c\gamma + p(w), \quad Q = AC^2 = w\gamma + q(w). \quad (5)$$

Then

$$\det \frac{\partial(P, Q)}{\partial(w, \gamma)} = -c\gamma, \quad \det \frac{\partial(w, \gamma)}{\partial(v, t)} = b\gamma.$$

Moreover,

$$\det \frac{\partial(x, v, t)}{\partial(x, y, z)} = x^3, \quad \det \frac{\partial(A, B, C)}{\partial(P, Q, C)} = -C^{-3}.$$

Multiplication, using $C = x\gamma$, gives

$$\det JF_p = bc \neq 0. \quad (6)$$

Thus F_p is a Keller map and A, B, C are algebraically independent.

3 Inverse equation and exact recovery

Equation (4) is equivalent to

$$\Phi(w) = wp(w) - cq(w).$$

Eliminating γ from (5) yields

$$H(T) := \Phi(T) - PT + cQ = 0. \quad (7)$$

Conversely, a generic root w of (7) recovers the source:

$$\begin{aligned} \gamma &= \frac{P - p(w)}{c}, & x &= \frac{C}{\gamma}, & u &= \frac{w}{\gamma}, & v &= u - 1, \\ y &= \frac{v}{x}, & t &= \frac{\gamma - 1 - av}{b}, & z &= \frac{t}{x^2}. \end{aligned} \quad (8)$$

There is a particularly short certificate against an extraneous branch:

$$H'(w) = p(w) - P = -c\gamma. \quad (9)$$

A root with $\gamma = 0$ would therefore be a repeated generic root. As proved immediately below, H is irreducible and hence separable in characteristic zero. Moreover, $C \neq 0$ on a generic open set, so all denominators in (8) are valid at the generic point.

The target change is birational:

$$\mathbb{C}(A, B, C) = \mathbb{C}(P, Q, C),$$

since $P = BC$, $Q = AC^2$, $B = P/C$, and $A = Q/C^2$. If $K = \mathbb{C}(A, B, C)$ and $L = \mathbb{C}(x, y, z)$, exact recovery gives

$$L = K(w). \quad (10)$$

If $\deg p = d - 1$, then $\deg \Phi = d$. The polynomial $\Phi(T) - PT + cQ$ is irreducible over $\mathbb{C}(P, Q)$: it is irreducible in $\mathbb{C}[P, Q, T]$ because it is primitive and linear in Q with unit coefficient, and it is primitive as a polynomial in T . Gauss's lemma applies. Adjoining the independent transcendental C preserves irreducibility. Hence (10) has degree d .

4 Full symmetric monodromy

We use the following classical theorem.

Theorem 3 (Brink [1, Theorem 13]). *Let k be a field whose characteristic does not divide $d(d-1)$. For fixed $a_{d-1}, \dots, a_2 \in k$, the polynomial*

$$X^d + a_{d-1}X^{d-1} + \dots + a_2X^2 + SX + R$$

has Galois group S_d over $k(S, R)$.

Divide (7) by the nonzero leading coefficient of Φ . Because $p(0) = 0$, the fixed polynomial Φ has no linear term. The linear and constant coefficients of the normalized equation are independent nonzero scalar multiples of P and Q . Taking $k = \mathbb{C}$ in Theorem 3 gives

$$\text{Gal}(H/\mathbb{C}(P, Q)) = S_d. \quad (11)$$

The field $\mathbb{C}$ is algebraically closed, so this is already the geometric group. Adjoining the independent transcendental C does not change the splitting-field group. Finally, (10) identifies its natural root action with the action on the d points of the generic fibre. Therefore $\text{Mon}(F_p) = S_d$, proving Theorem 1.

Gallagher's construction supplies admissible seeds in every degree $d - 1 \geq 2$. Since $d > 1$, the associated map cannot be a polynomial automorphism. This proves Corollary 2.

5 A compact rational family

One particularly simple choice, useful for direct checking, is

$$b = c = 1, \quad p_d(w) = \frac{2w - dw^{d-1}}{d-2}.$$

It satisfies (1), has $\kappa = -(d+1)$, and gives

$$a = -\frac{d}{d-1}, \quad q_d(w) = \frac{w^2 - (d-1)w^d}{d-2}.$$

Thus

$$u = 1 + xy, \quad \gamma = 1 - \frac{d}{d-1}xy + x^2z$$

and

$$\begin{aligned} A_d &= \frac{(d-2)u + u^2 - (d-1)u^d\gamma^{d-2}}{(d-2)x^2}, \\ B_d &= \frac{(d-2) + 2u - du^{d-1}\gamma^{d-2}}{(d-2)x}, \quad C_d = x\gamma. \end{aligned}$$

After $U = (d-2)P$ and $V = -(d-2)Q$, the inverse polynomial is simply

$$T^d - T^2 + UT + V.$$

This is also a direct instance of Brink's theorem. Alternatively, $T^d - T^2 + UT$ is Morse over $\overline{\mathbb{C}(U)}$: its critical points are simple, and equality of two critical values with ratio r would imply

$$(d-2)r^d - dr^{d-1} + dr - (d-2) = 0,$$

forcing the transcendental parameter U to be algebraic. The classical Morse-polynomial theorem [5, Theorem 4.4.5] again gives S_d .

There is an explicit collision. Put

$$s_d = \frac{4 - 2^d}{d-2}.$$

At the target $(A_d, B_d, C_d) = (s_d, s_d, 1)$, the inverse polynomial has the distinct roots 1 and 2. Their recovery denominators are

$$\gamma(1) = \frac{d+2-2^d}{d-2} \neq 0, \quad \gamma(2) = 2^{d-1} \neq 0.$$

Hence every displayed map is visibly noninjective.

6 Priority and verification

Gallagher [2] proved the weighted-lift construction and the occurrence of every generic degree at least three. Brink [1] proved the Galois theorem used here. Before this record, MathOverflow answer 513470 and its linked Note 19 [3, 4] proved S_d for Gallagher’s canonical tower for $3 \leq d \leq 13$ and reported further finite checks. We claim none of those ingredients. A source-specific search found no earlier statement that Brink’s theorem makes *every* admissible Gallagher seed symmetric in every degree. Absence of a search hit is not proof of worldwide priority.

The accompanying exact verification has three parts. A symbolic script keeps all unconstrained seed coefficients symbolic and checks the endpoint, integral, divisibility, elimination, and recovery identities for $3 \leq d \leq 10$. A separate PARI/GP resultant calculation verifies at the specialization $U = 1$ that all $d - 1$ finite branch values are simple for $3 \leq d \leq 20$, independently checking the geometric Morse mechanism. Exact number-field specializations give arithmetic S_d for $3 \leq d \leq 10$; these are only arithmetic consistency checks and are not presented as independent proof of geometric monodromy.

AI-assistance disclosure. This research record, proof organization, source search, and verification code were produced with substantial assistance from OpenAI Codex agents. The primary human researcher directed the program. No person was contacted by the agents. The theorem passed an internal hostile-agent audit, which is not peer review.

References

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